Insurance Costs – Analysis of Factors Related to Costs

Isabel Holm

2026-04-26

Introduction

This report presents an analysis of insurance cost data with the aim of identifying which factors are most strongly associated with individual insurance charges. The dataset contains information on customer demographics, lifestyle factors, health-related variables, and insurance history.

The purpose of this analysis is twofold: first, to explore and describe patterns in the data through statistical summaries and visualizations; and second, to build regression models that can help explain variation in insurance costs and potentially serve as a support tool in future pricing decisions.

The data set contains information about **customer_id, age, sex, region, bmi, children, smoker, chronic_condition, exercise_level, plan_type, prior_accidents, prior_claims, annual_checkups** and **charges**.

Method

The analysis was conducted in R and structured across three main stages: data preparation, descriptive analysis, and regression modeling.

Data Preparations

The data set was loaded using the `readr` package and initially inspected using `glimpse()` and `summary()` to understand its structure and variable types.

Missing values were identified in three variables: **bmi** (28 missing), **exercise_level** (22 missing), and **annual_checkups** (20 missing). Missing values were handled individually with the planned regression analysis in mind. For each variable containing missing observations, a

missingness indicator variable was created, followed by imputation. Missing values in **bmi** and **annual_checkups** were imputed using the median, while missing values in **exercise_level** were replaced with *"Unknown"*.

Further, categorical variables were standardized by trimming whitespace and applying consistent title case formatting, before being converted to factors. A new ordered variable, `age_group`, was created by binning age into five intervals (18–30, 31–40, 41–50, 51–60, 61–75) to support group-level comparisons in the descriptive analysis. In addition, a derived variable, `risk_group`, was created to capture combined risk based on prior insurance history. This variable combines *prior_claims* and *prior_accidents* into a single categorical measure. Policyholders with no prior claims or accidents were classified as **Low** risk, those with two or more claims or accidents as **High** risk, and all others as **Medium** risk.

Descriptive Analysis

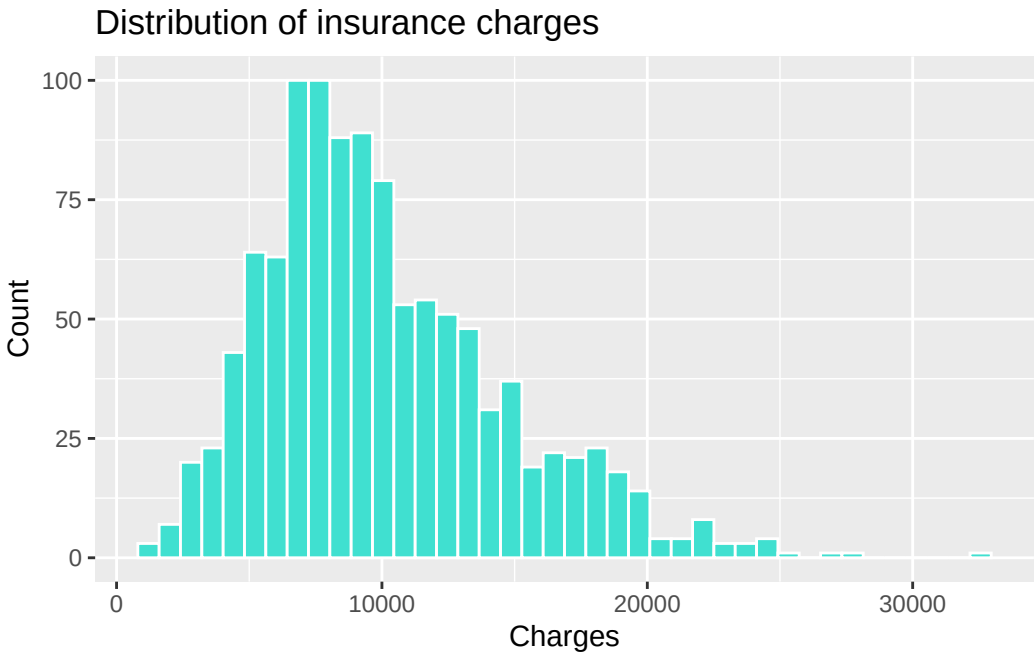
To get a better understanding of the data, a descriptive analysis was conducted. Summary statistics for the target variable **charges** were computed, including mean, median, standard deviation, min, max, and interquartile range. A range of visualizations were also produced using `ggplot2` to explore the distribution of charges and its relationship with key variables. In the result section, only the visuals showing the strongest predictors will be displayed and discussed. However, the other variables that are not among the strongest predictors on their own may still contribute to the regression models.

Statistical summary

Statistic	Value
n_charges	1100
mean_charges	10060
median_charges	9124
sd_charges	4574
min_charges	1204
max_charges	32559
iq_r_charges	5873

The dataset contains 1,100 observations. Insurance charges range from 1,204 to 32,559, with a mean of 10,060 and a median of 9,124. The gap between the mean and median suggests a right-skewed distribution, where a smaller number of high-charge customers pull the average upward. The standard deviation of 4,574 and an interquartile range of 5,873 indicate considerable spread in charges across the dataset.

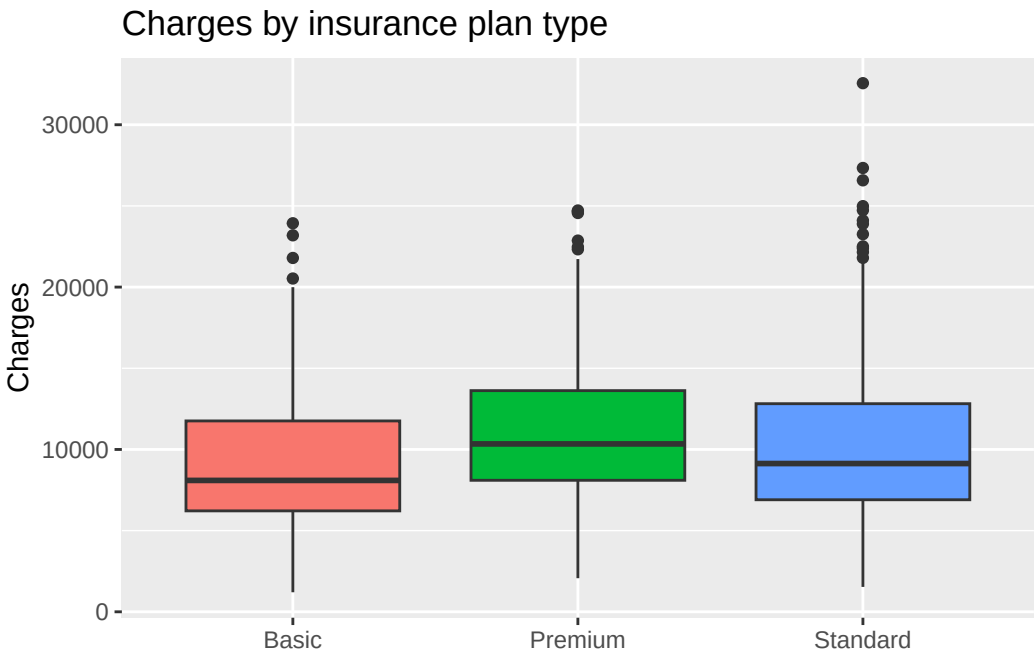
Distribution of charges



The histogram confirms the right-skewed distribution noted in the summary statistics. The majority of customers have charges clustered between approximately 4,000 and 13,000, with a long tail extending toward the higher end. This suggests that while most customers incur moderate costs, a smaller group generates significantly higher charges.

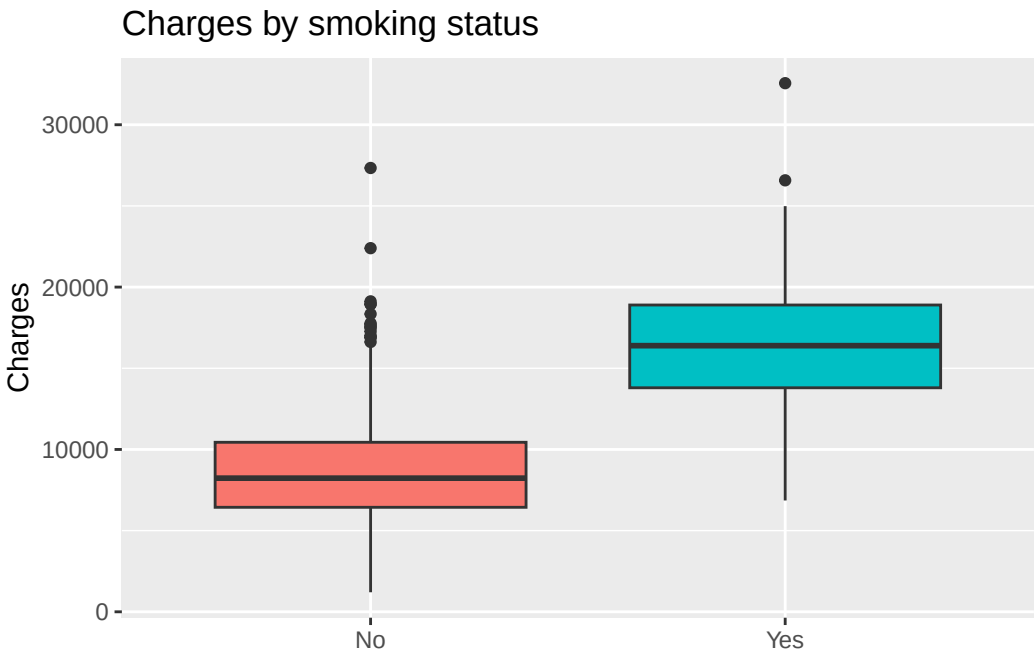
Charges per group

Charges by insurance plan type



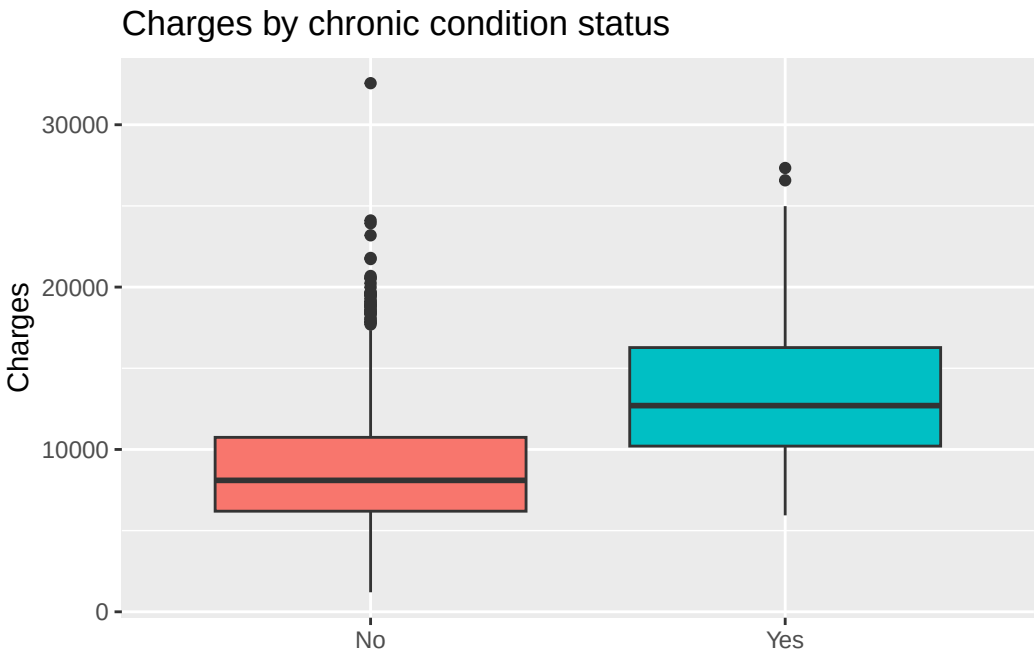
Charges by insurance plan type show that Premium plan customers tend to have somewhat higher median charges compared to Basic and Standard customers. The Standard plan displays the widest spread and the most outliers at the upper end, suggesting greater variability in costs among that group. The difference between plan types is moderate, indicating that plan type alone may not be a strong standalone predictor of charges.

Charges by smoking status



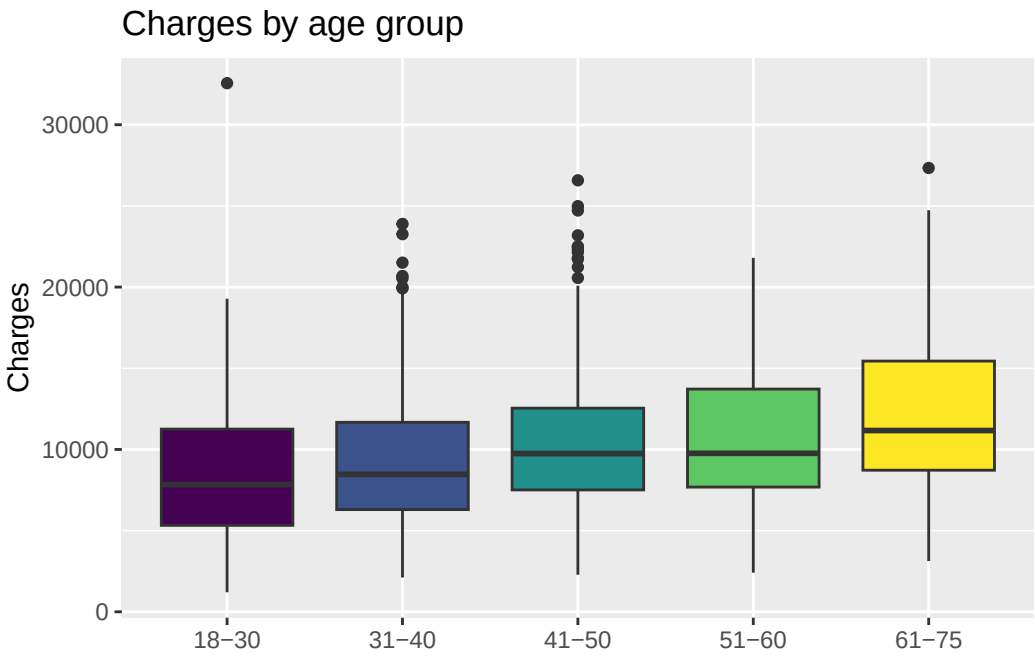
Smoking status shows one of the clearest group differences in the dataset. Smokers have a considerably higher median charge of around 16,000 compared to approximately 8,000 for non-smokers. The interquartile range for smokers also sits entirely above the median for non-smokers, indicating that even typical smoker costs exceed what most non-smokers pay. This suggests smoking is likely to be one of the stronger predictors in the regression model.

Charges by chronic condition



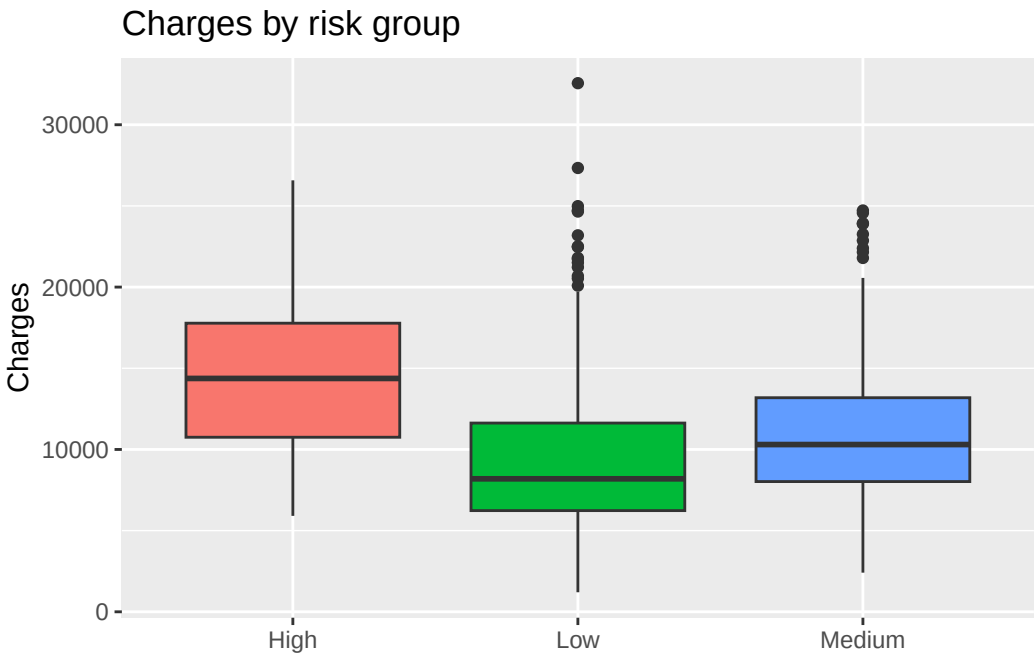
Customers with a chronic condition show notably higher charges than those without, with a median of around 13,000 compared to approximately 8,000 for the non-chronic group. The spread among chronic condition customers is also tighter, suggesting a more consistent cost pattern within that group. This points to chronic condition status as another likely significant predictor in the regression model.

Charges by age group



Charges show a clear upward trend across age groups, with median charges rising steadily from around 8,000 in the 18–30 group to approximately 12,000 in the 61-75 group. The spread also widens in older groups, particularly in the 61-75 bracket, suggesting greater variability in costs among older customers. This pattern aligns with the expectation that age is positively associated with insurance charges and indicates it will likely be a meaningful predictor in the regression model.

Charges by risk group

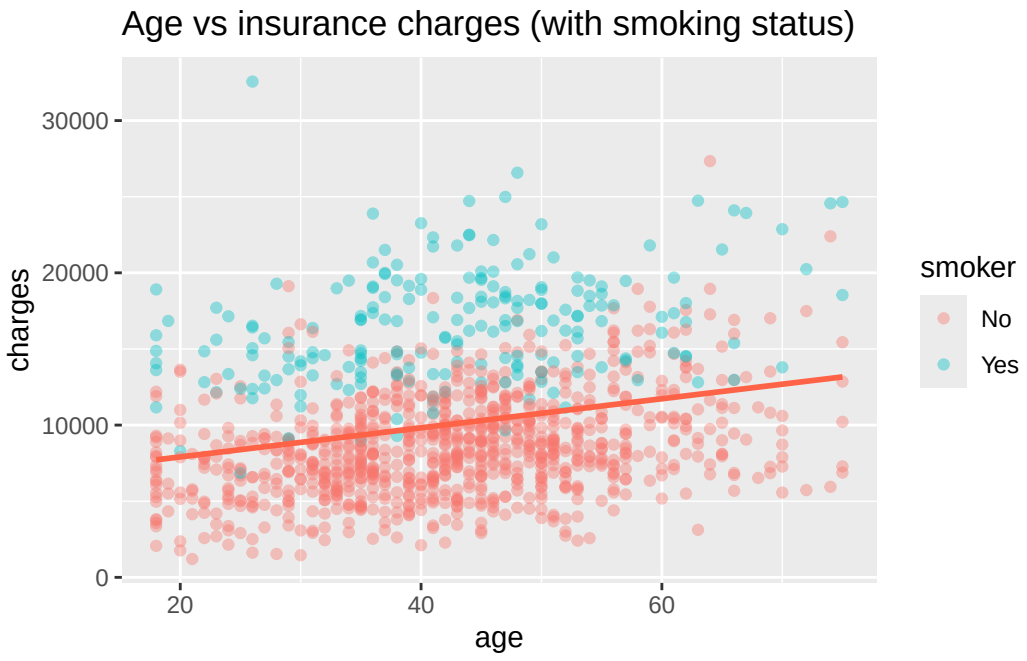


Customers were grouped into three risk categories based on their claims and accident history. The boxplot shows a clear stepwise pattern, with **High** risk customers having a median charge of around 14,000, followed by **Medium** at approximately 10,000 and **Low** at around 8,000. This suggests that prior claims and accident history combined into a single risk group variable is a meaningful way to capture customer risk and is likely to be a relevant predictor in the regression model.

It should be noted however that the thresholds used to define the groups are manually set and somewhat arbitrary, different cutoff values could produce different groupings and potentially lead to different conclusions.

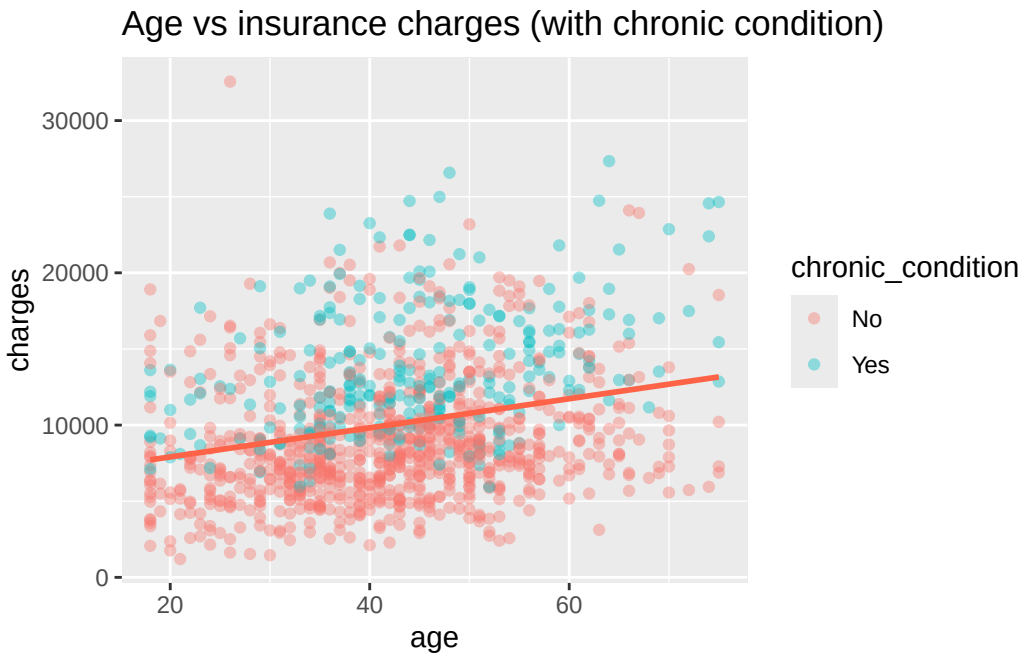
Continuous predictors and charges

Age against charges with smoking status



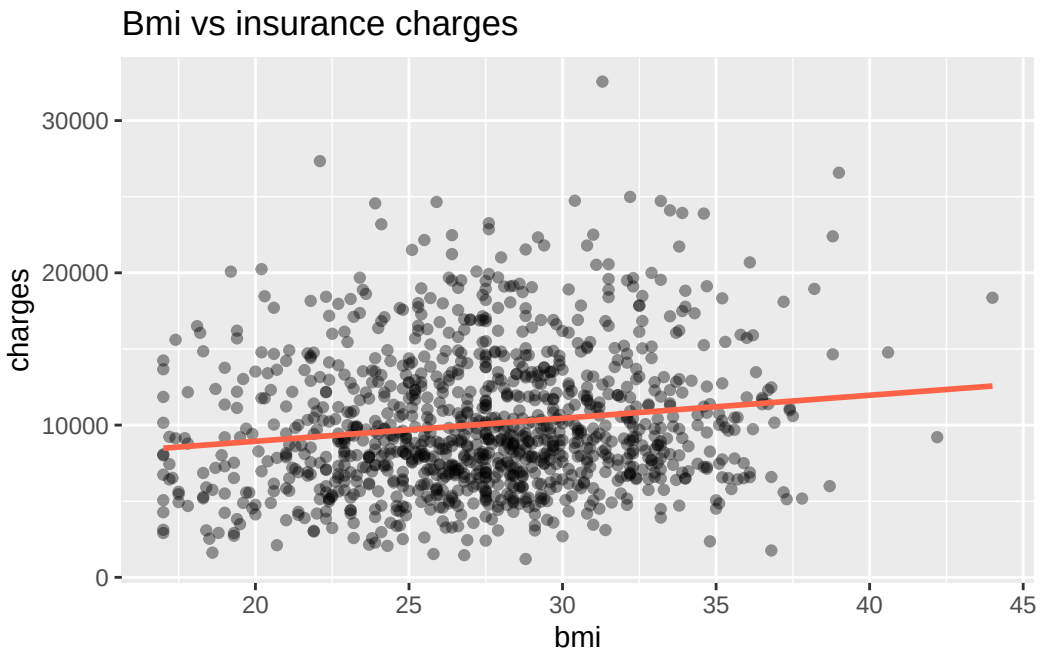
The scatter plot of age against charges, colored by smoking status, reveals two fairly distinct clusters. Smokers consistently occupy the upper portion of the chart across all age groups, while non-smokers cluster at lower charge levels. The overall trend line shows a positive relationship between age and charges, but the separation between smokers and non-smokers suggests that smoking status may have a stronger influence on charges than age alone.

Age against charges with chronic condition



When grouping by chronic condition, the separation between groups is less pronounced than with smoking status. Customers with a chronic condition tend to have somewhat higher charges, but there is considerable overlap between the two groups. The positive relationship between age and charges holds regardless of chronic condition status.

Bmi against charges



BMI against charges shows a weaker positive relationship, with the trend line indicating that charges increase slightly as BMI rises. However, the considerable spread of points around the trend line suggests that BMI alone is a relatively weak predictor of charges. It may still contribute to the regression model, but is unlikely to be among the strongest predictors on its own.

Regression analysis

To predict insurance charges, three linear regression models were built using `lm()`. The models were constructed progressively, starting simple and adding complexity stepwise, in order to evaluate whether additional variables and interactions improved predictive performance.

The descriptive analysis indicated that three variables had a particularly strong relationship with charges: `smoker`, `chronic_condition`, and `age`. These were included in all three models as the core predictors.

- **Model 1:** includes only these three strong predictors and serves as a baseline for comparison.
- **Model 2:** extends the baseline by adding `bmi`, `risk_group`, `plan_type`, and `exercise_level`. These variables were selected based on their theoretical relevance to insurance costs, and although their relationship with charges was weaker than the core predictors, the descriptive analysis indicated a positive association worth exploring further.

- **Model 3:** builds on Model 2 by adding an interaction term between `age` and `smoker`, based on the hypothesis that the cost of smoking may increase with age. An interaction term lets the model capture that.

Variables such as `sex`, `region`, and `annual_checkups` were excluded to keep the model interpretable, and because their theoretical link to individual charges was considered weaker or more ambiguous.

Model comparison

The three models are compared against each other using adjusted R^2 and AIC (Akaike Information Criterion). Adjusted R^2 measures how well the model explains the variation in insurance charges, expressed as a value between 0 and 1 where a higher value indicates a better fit. Unlike regular R^2, it adjusts for the number of predictors, meaning it only increases if a new variable actually improves the model.

AIC balances how well the model is fitted against model complexity, where a lower value indicates a better model. It is particularly useful when comparing models of different sizes, as it penalises unnecessary complexity.

Table 2: Model comparision - Adjusted R^2 and AIC

Model	Adjusted R^2	AIC
Model 1	0.663	20474
Model 2	0.741	20192
Model 3	0.741	20194

Adjusted R^2: Model 1 explains about 66% of the variation in insurance charges using only the three strong predictors. Model 2 jumps to 74% by adding BMI, risk group, plan type and exercise level, which is a meaningful improvement. Model 3 stays at 74%, meaning the interaction term between age and smoker didn't add any real explanatory value beyond model 2.

AIC: Model 1 has the highest AIC, confirming it's the weakest fit. Model 2 scores only slightly lower at 20 192 and Model 3 scores just above at 20 194, so once again the interaction term in Model 3 didn't improve the fit enough to justify it.

Overall conclusion: Model 2 is the best model. It explains noticeably more than Model 1, and the interaction between age and smoking that Model 3 tests turns out not to be necessary, at least not in this data set.

Model Conclusions

Model 2 - Detailed Results

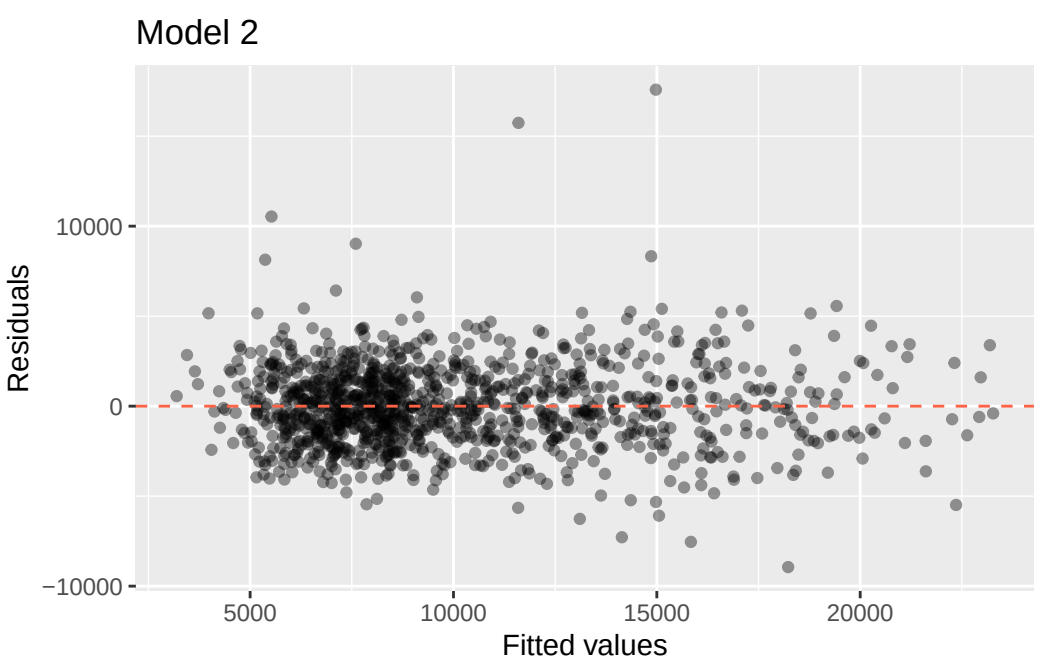
Model 2 explains 74% of the variation in insurance charges, and the overall model is highly significant ($p < 0.001$). All included predictors were statistically significant, suggesting that each variable contributes meaningfully to predicting charges.

The strongest individual predictor was *smoking status*, where smokers on average incur approximately 7 423 higher charges than non-smokers. Having a *chronic condition* adds around 3 791 to charges, and each additional year of age increases charges by approximately 71. BMI also shows a positive effect, with each unit increase adding around 159 to charges.

Regarding *risk group*, policyholders classified as Low or Medium risk have notably lower charges compared to the High risk reference group, with Low risk customers having charges around 2 618 less. Customers on a Premium plan have charges around approximately 1 484 more than the Basic plan, and those with a Standard plan around 670 more. Finally, *exercise level* shows that lower activity is associated with higher charges, with Low exercise adding around 1 488 compared to the High exercise group.

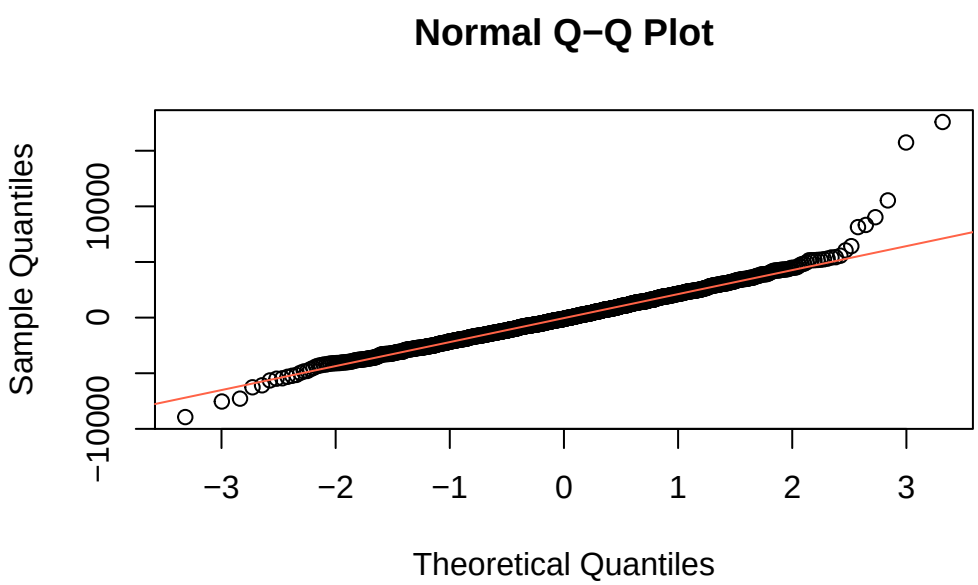
Note that in all interpretations are made on the basis that each coefficient reflects the effect of that variable while holding all others constant.

Residual vs fitted



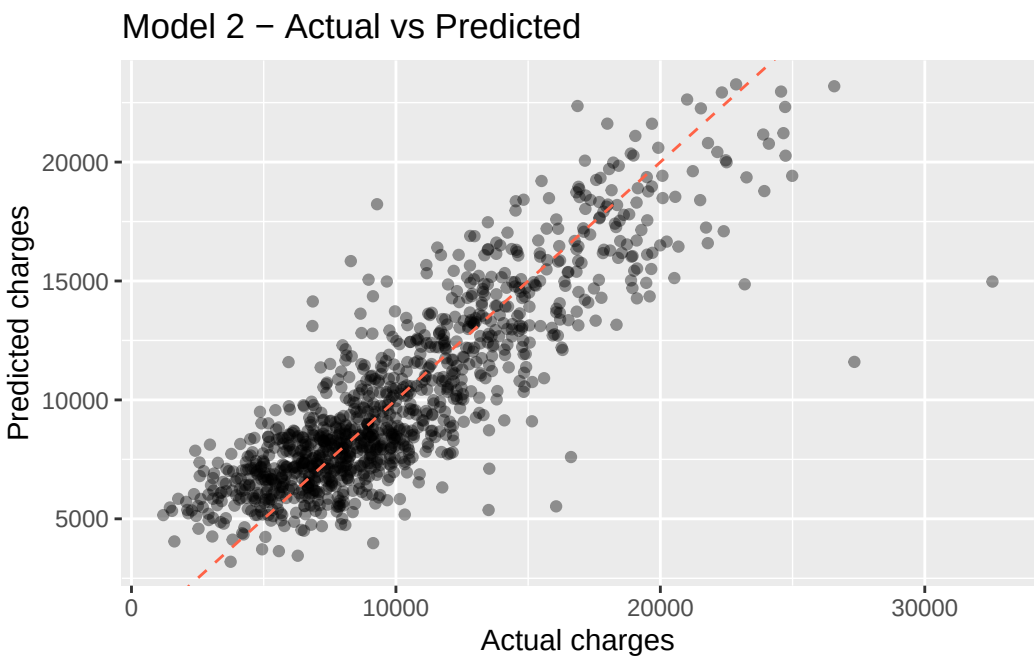
The residuals are roughly centered around zero, which is a good sign. However, the spread increases as fitted values get larger.This means the model is less precise at predicting higher charges, which is worth noting as a limitation. A few outliers with large residuals are also visible, particularly in the mid-to-high range.

QQ-plot



The residuals follow the reference line closely through the middle of the distribution, suggesting approximate normality for most observations. However, both tails deviate from the line. This indicates that the model tends to underestimate some very low charges and overestimate some very high ones.

Actual vs predicted charges



Points close to the dashed reference line indicate accurate predictions. The model performs well in the low-to-mid charge range, where most points follow the line closely. However, at higher actual charge levels the predictions become less accurate. The model tends to underpredict high charges, as seen by points falling below the reference line in the upper range.

A few notable outliers are also visible, particularly one observation with very high actual charges, about 32 000, that the model significantly underpredicts. Overall the plot confirms that the model captures the general pattern reasonably well, but is less reliable at the extremes.

Limitations

1. Linear regression assumes that the relationship between each predictor and charges is constant and straight. For example, that each additional year of age always adds the same amount of charges. However, in reality this might not be the case. The effect of relationship could accelerate over time or the effect of an relationship might only become relevant over a certain threshold. A linear model will fail to fully capture non-linear relationships, which can lead to systematic prediction errors.
2. Linear regression assumes that the residuals are normally distributed. When charges are skewed, in which was visible in the QQ-plot where tails deviated from the reference line, the model's predictions and statistical tests become less reliable, particularly at the extremes of the distribution.

3. The model is built on historical data and can only identify which variables are associated with higher charges, but it cannot tell us why. For example, the model shows that smokers have higher charges, but cannot confirm that smoking directly causes the higher costs.
4. Variables such as `sex`, `region`, `annual_checkups`, and `children` were left out of the models to keep them more simple. While some of these may have a weak relationship with charges individually, they could still contribute when combined with other variables. A more comprehensive model might include them and potentially explain a larger share of the variation in charges.
5. `risk_group` is a simplified derived variable that loses some nuance from the raw claims/accidents data. While combining two variables may make them easier to interpret, it comes with a cost of detail. Using the raw variables separately, or a more granular scoring approach, might capture risk more accurately.

Self-reflection

I am satisfied with the structured approach I took throughout the project, primarily for the regression analysis, starting with a simple baseline model and building further on it. I also enjoyed learning how to use Quarto to build my report. The most challenging part was the coding, I usually have a clear idea of what I want to do analytically, but translating that into working R code remains the hardest part for me.

I believe this submission meets the VG criteria. I fulfill all baseline requirements for Godkänd, and go beyond them by comparing multiple regression models, motivating model choices, and critically discussing model fit, limitations and possible improvements, which I see as aligned with the deeper level of understanding the VG grade requires.